

Privilege Escalation Methods (Medium–Hard Boxes)

1. PATH Hijacking

- Sudo/script uses binaries without full path.
- Create malicious binary in writeable directory.
- Modify \$PATH or exploit existing PATH search order.

2. Environment Variable Abuse

- env_reset disabled in sudoers.
- Exploit LD_PRELOAD, LD_LIBRARY_PATH, PYTHONPATH, SHELL, EDITOR.

3. Python Library Hijacking

- Sudo runs Python script.
- Script imports local modules.
- Create malicious module in writeable import path.

4. Wildcard Injection

- Abusable commands: tar, rsync, cp, chmod.
- Create --checkpoint-action files to trigger root execution.

5. SUID Binary Abuse

- Find SUID: find / -perm -4000.
- PATH hijack, unsafe system() calls, format strings, overflows, capabilities.

6. sudoedit / Symlink Attacks

- sudoedit allowed on editable targets.
- Replace target with symlink to root file.

7. Systemd Service Misconfig

- Modify or override service files.
- Add malicious ExecStart and restart.

8. Docker / LXC Breakouts

- User in docker/lxc group.
- Run container mounting / and chroot.

9. Cronjob Abuse

- Writable scripts or paths used in cron.
- PATH hijack or wildcard injection.

10. Restricted Shell Escapes

- Escape rbash via vim, less, awk, find, python, etc.

11. Kernel Exploits

- Outdated kernel vulnerable to DirtyPipe, DirtyCow, OverlayFS, Netfilter.

12. Abusing Privileged Services

- Insecure UNIX sockets, DBus, monitored folders.

13. Regex/Filter Bypass

- Inject commands bypassing weak regex filters.

14. Log Poisoning + LFI

- Writable logs + LFI → code execution.